

MUHAMMAD HAIDER ZAIDI

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Technical Skills

Programming: Python, C++, SQL, Javascript

AI/ML Frameworks & Agents: PyTorch, TensorFlow, LLMs, OpenAI API, AI Agent Orchestration

MLOps: Docker, Kubernetes, AWS SageMaker, FastAPI, CI/CD, Git, Production Systems, API Deployment

Data Engineering: PySpark, Airflow, ETL Pipelines, System Integration, Data Architecture, API Design

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and evaluated **deep learning models** and architectures for metrological applications, optimizing performance and training efficiency.
- Explored novel **AI architectures** and optimized models to enhance accuracy and stability in complex systems.
- Collaborated on research contributing to **AI agent development** and published findings, ensuring robust technical decisions.
- Developed dashboards for **production system monitoring** and model performance analysis, aiding continuous improvement.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Optimized **AI algorithms** for vitiligo segmentation, achieving 94% accuracy and 5x speed increase; deployed via **Docker and FastAPI** ensuring **production stability**.
- Engineered a **real-time AI agent** for tennis foul detection, leveraging TensorRT to reduce compute cost and time by 6x, demonstrating efficient **backend architecture**.
- Managed **MLOps pipelines** on AWS SageMaker, addressing model data drift and maintaining **Kubernetes infrastructure**, ensuring robust **CI/CD** and **production stability** for **backend services**.
- Developed proof-of-concept **AI solutions** (YOLO, few-shot learning, CNN) for client demonstrations, showcasing versatile **platform components**.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Full-stack development of a web platform offering **AI-based services**, focusing on robust **backend services** and **API design** using Flask.

- Developed and integrated an **AI chatbot application** using the OpenAI API, demonstrating expertise in **AI agent orchestration** and **API integration**.

Hugging Face

Oct 2022 – Nov 2022

Open-Source Contributor

Remote

- Co-authored technical documentation and implementation for Zero-shot Image Classification on Hugging Face Hub, contributing to **AI platform components**.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Designed and implemented **ETL pipelines** and web scrapers for structured and unstructured data, ensuring efficient **system integration**.
- Automated data ingestion and cleaning workflows, enhancing data quality for client analytics and supporting **integration processes**.

Systems Ltd. - Visionet

June 2021 – Sep 2021

Data Engineer Intern

Lahore, PK

- Gained hands-on experience with the Hadoop ecosystem (Hive, Sqoop, NiFi) and advanced **ETL processes** using PySpark and Airflow.
- Authored technical content on data engineering concepts, facilitating knowledge transfer in **data architecture** and **integration**.

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Architected a weighted-probability ensemble integrating diverse models, leveraging AlphaFold for feature extraction to enhance **AI agent** accuracy and stability.
- Achieved a 4% increase in **AI model** accuracy through custom feature engineering, demonstrating expertise in **data processing** and **backend architecture**.

Razanlytics: Real-time Bitcoin Price Forecasting

2022

Thesis Project

- Developed a real-time Bitcoin price forecasting system, integrating diverse data sources and deploying predictive **AI models** through **APIs**.
- Ensured **production stability** for data streams and model inference, utilizing robust **backend services** for high availability.

Saaz: AI Self-Playing Chitralli Sitar

2023

Signal Processing, Music Generation

- Designed **real-time backend services** for signal processing, integrating temporal dependencies for music generation and **AI agent control**.
- Implemented transformer-based **AI models** for sequence prediction in audio generation, demonstrating advanced **AI agent development**.

Hugging Face

Open-Source

Contributor

- Co-authored Zero-shot Image Classification documentation and implementation page on the Hugging Face Hub, contributing to **AI platform components** and community knowledge.

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Specialized in deep learning for computer vision and spatio-temporal modeling, relevant for **AI agent development**.
- Relevant coursework: Advanced Neural Networks, Computer Vision, Autonomous Systems, focusing on **backend architecture** and **system integration**.

National University of Computer and Emerging Sciences Aug 2018 – June 2022

Bachelor of Science in Computer Science

Karachi, PK

- Dean's List of Honor
- Specialized in Machine Learning and Computer Vision with focus on time-series analysis, providing foundation for **AI software engineering**.

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python